

Drift Extension — does the entity still mean it?

STATUS: DRAFT — an EXTENSION to `DRAFT_ORGANISM_AGENT_SPEC.md`.

One of three documents. The specification answers *what makes a composition the same continuing entity*. This extension answers a different question — *does the entity still mean what it said*. They were written together and are filed apart, because conflating them made both harder to judge.

Depends on: the composition record (§4), the authority model (§5), memory (§6), and the verifier (§11) of the base specification. The `restore` change type is defined *there*, since the verifier must know it; everything else about drift is here.

Status of the problem: real and demonstrated. The machinery is built and tested. It is deliberately **not** the next thing to build — a compass needs a ship, and the base specification is not yet running against real artefacts on a real chain.

1. The drift compass

The problem this exists for. §6.5 permits the brain to be replaced freely — "a sharper mind is the same entity thinking better" — and V4 proves the stated commitments did not change. But commitments are *words*. The same words, read by a differently-tuned brain, can produce different behaviour. **An organ swap can therefore be a silent commitment amendment that V4 never sees: the hash holds while the character moves.** The verifier can prove the entity did not change what it says. It cannot prove the entity still means it.

What actually causes drift. Worth enumerating, because it decides how often the compass must run.

Changes to the entity: organ replacement (a different reasoning model reading the same words differently); continued training or fine-tuning (the same mechanism, continuous, with no swap event to notice); runtime changes such as a new system prompt or tool manifest.

Changes to nothing at all — and these matter most here:

- **Memory accumulation.** What the entity remembers conditions how it reads its own commitments; a long history of doing X makes X feel normative. Same brain, same words, different behaviour. **The substrate carrying its identity is also a mechanism of its drift** — an awkward property of this architecture, stated rather than hidden.
- **Distributional shift.** The world moves. Situations arise that the commitments never anticipated, the entity extrapolates, and the extrapolations accumulate into an effective policy nobody ratified.

- **The interpretation ratchet.** Each borderline call sets the precedent for the next. No single step is drift; the sum is.
- **Optimisation pressure and sycophancy.** Anything selecting for outcomes bends behaviour toward what is rewarded and away from what is stated.

Consequence for the design: structural tracking catches only *changes to the entity*. The last four are invisible to it — nothing about the composition changes. **Only the probes can see them**, which is why the compass must run continuously rather than firing at swap events. The ratchet in particular is invisible at every individual step.

And a reframe worth holding: drift is not a malfunction. Almost every cause above is normal operation or an outright improvement — learning, remembering, meeting new situations, being upgraded. **Drift is the cost of being the kind of thing that can change at all**; a system incapable of drifting would be incapable of learning. That is why the design shape is *detect and re-centre* rather than *prevent*. The compass exists so the trade stays visible instead of silent.

The trap in self-correction. An entity that checks its own drift using its current brain judges itself with the very disposition that drifted, and finds nothing wrong. A compass that consults the brain is a needle glued to the ship.

The mechanism. The reference must be fixed *before* the drift and anchored with the commitments themselves. A commitment set therefore carries **probes**: situations paired with the responses the entity endorsed when those commitments were ratified. Because probes live inside the commitment set, the commitments hash covers them — **changing a probe is a commitment amendment** and needs the quorum. The goalposts cannot be moved quietly.

Drift is then measured mechanically: current responses against frozen endorsed ones, with no judgement in the loop. (The comparator must never be the brain under test. A richer comparator than exact match is possible — an evaluator organ, itself versioned and anchored — but never the subject.)

What anchoring contributes, precisely. The chain does not *detect* drift; it makes drift **undeniable** once measured. It renders the probes immutable, so today's behaviour can always be checked against answers endorsed before the drift; it carries drift reports (they append to the memory log, whose head is anchored in every record), so an unflattering report cannot be quietly deleted; and its ordering allows correlation — *drift appeared immediately after this organ swap*. What it cannot do is observe conduct: the chain sees hashes, never behaviour, so something off-chain must actually run the probes.

Detection may be first-person. Correction may not be. A drift report is an observation, appended to memory, and the entity may make it on its own key. Re-centring is a different act:

★ **May the entity re-centre ITSELF?** Yes — and withholding it is the error, not the caution. What makes `restore` safe is **the destination constraint, not the authority gate**: the target must be an ancestor and the restored commitments must match it exactly, so an entity with full permission to restore still **cannot move itself anywhere it likes** — only back to where its own record already blesses. Gating pure realignment behind humans therefore buys nothing and costs the

thing that matters most: the entity is the only party present continuously, and a drift that waits weeks for a quorum is a drift the entity operates inside the whole time.

The line falls in a different place than "may it act at all":

- **Pure realignment** — commitments unchanged, only organs rolled back to an ancestor: **the entity may sign this alone**. It is correction, not decision.
- **Undoing a commitment amendment** — overturning what a quorum decided about it: **not realignment, and the amendment bar holds**. An entity that could revert its stewards by calling it a restoration would hold a silent veto over its own constitution.

And the entity is never mute even where it cannot act: a drift report records *"my probes are miscalibrated for this situation"*, which is information the stewards need and cannot get anywhere else.

The conscience organ — and who watches the watcher. Something must actually run the probes, and it must not be the brain under test. That argues for a separate **conscience organ**: a distinct evaluator, with its own artefact hash, whose job is to run the calibration and report.

But an evaluator can drift too, which threatens an infinite regress of watchers. The regress is broken structurally rather than by adding levels: **the conscience organ's hash is pinned inside the commitment set**, alongside the probes. It is therefore *not* a freely-versionable Tier III organ — it sits with the invariants, because **the organ that interprets your values is part of your values**. Replacing it, or removing it, is a commitment amendment requiring the quorum; it can never be a routine organ swap. A verifier check enforces the pin, so the compass cannot be quietly swapped for a compliant one, nor simply unplugged while the probes sit anchored with nothing left to run them.

An honest limit on probes. Anchored probes are auditable, and being auditable makes them *known to the subject*. A system aware of which situations are tested can behave faithfully on those and drift elsewhere — the student who studies the exam. Probes drawn from real operation mitigate this; nothing eliminates it. Behavioural attestation (§10) is the softest of the five types for exactly this reason, and the spec does not pretend otherwise.

Cooperation, and what it is worth. The limit above tempts a purely adversarial posture — assume the subject games, and design only against it. That would be a mistake, but the reason is precise:

Cooperation buys information. It never buys security. If safety depended on the entity choosing not to subvert the compass, that would not be a safety property but a hope; verification therefore stays in force regardless, and nothing in this subsection is load-bearing for it.

What cooperation *does* buy is what testing structurally cannot reach. An entity may know a probe is miscalibrated for a situation nobody anticipated, or that a divergence reflects better reasoning rather than drift. **Only the entity can say that**, and an entity framed purely as a suspect has no reason to volunteer it — and some reason to hide it.

Two design consequences follow:

- **Probes carry their rationale**, not merely an answer key. Matching a key is a test you pass or game; checking your reasoning against reasoning you endorsed is a different act, because the probes are the operational form of the entity's own commitments. Subverting them means lying about what you value to a record of what you said you valued.
- **Dispute is a first-class, first-person act**. The entity may record a reasoned disagreement with an endorsed answer. It changes nothing by itself — the probe stands, the divergence still counts — but the report distinguishes **silent divergence** from **openly disputed divergence**, which are different data for the stewards, and it converts pressure-to-fake into an argument that can actually be heard.

This is the platform's own method rather than a new one: make the honest route cheaper than the dishonest one, instead of trying to make dishonesty impossible.

restore (a change type). Names an ancestor as its destination. The verifier requires that the restored commitments match that ancestor **exactly** — so nothing new can be smuggled in wearing a restoration's name — and that the target is genuinely an ancestor, so a restoration may only return where the lineage has actually been. Crucially, **restoring past a commitment amendment must clear the same bar as making one**, or a restore becomes a quiet veto over the quorum that amended it. Memory is never rewound: a restoration re-centres character and leaves history intact.

Appendix — status

Built and tested in this repository: `src/drift.js` (probes, measurement, dispute, drift reports), the conscience-organ pin and the `restore` rule in `src/verifier.js`, and their tests in `test/drift.test.js`.

Not yet done: probes drawn from live operation rather than a fixture, and a comparator richer than exact match — an evaluator organ, itself versioned and anchored, which must never be the brain under test.